

Chapter 7

Reducing the Risk of Low Back Injury

As I have suggested in previous chapters, many of the classic instructions from experts about lifting (e.g., bend the knees and not the back) are in most cases neither feasible nor justifiable. In fact, very few occupational lifting tasks can be performed this way. The resulting tragedy is that clinicians or other experts lose credibility in the eyes of the worker; the worker knows she cannot do her job that way. She lifts objects from the floor, out of parts bins, over tables, and so forth. The science shows that squatting can have an increased physiological cost and that, depending on the characteristics of the load, it may not even reduce loads. Squatting is not always the best choice of position; it depends on the specifics of the task (e.g., the size, weight, and density of the object; the pick and place location; the number of repetitions). For example, the golfer's lift (see [figure 7.6](#) later in this chapter) may be preferable for reducing spine loads to bending the knees and keeping the back straight for someone repeatedly lifting light objects from the floor.

Reducing pain and improving function for patients with low back pain involve two components:

- Removing the stressors that create or exacerbate damage
- Enhancing activities that build healthy supportive tissues

This chapter addresses the issue of prevention—specifically, reducing the overloading stressors that cause occupational low back disorders (LBDs). After reviewing lessons from the literature and presenting the scientific issues, I provide two sets of guidelines: one for workers and another for employers. Because back belts have also been a much-debated topic and the subject of legislation in some jurisdictions, a reasonably in-depth discussion is presented. Finally, I offer a few notes to enhance the effectiveness of consultants. The lists provided in this chapter will result in more successful injury prevention programs. Practitioners and consultants who employ them will stand out from their peers.

Upon completion of this chapter, you will be able to formulate scientifically based guidelines for any activity that reduce the risk of

occupational low back disorders. Further, you will be aware of, and so be able to avoid, the pitfalls that cause ergonomic approaches to fail. Finally, you will be a more effective consultant by harnessing the real expertise—that of the worker.

Lessons From the Literature

Why does industry care about the backs of workers? Competitiveness in the new economy requires being profitable, and for many, the only way to enhance profit is to maximize loss control. A major source of losses to North American industry is worker injury, which results in direct costs for compensation and indirect costs of hiring and training replacement workers as well as reduced productivity because of decreased speed and increased errors. For this reason, preventing injury and promoting the rapid return to work of injured workers have become a major focus for industry. Because back injury represents an enormous cost in both real dollars and suffering, companies are realizing the benefits of supporting injury prevention and rehabilitation programs. Following are a number of studies that have addressed the issues that must be examined if such programs are to be successful.

Compensation Board Statistics—An Artifact?

When completing injury report forms, clinicians are often requested to name the event that caused the injury. These events are compiled by compensation boards to create their impressions of injury mechanisms—for example, the injured worker lifted and twisted. Although these named events may have been the culminating event, they were not the cause of the cumulative trauma that was present. As a result, the injurious event continues. Addressing the culminating event rather than the cause of the cumulative trauma will have little effect.

Ergonomic Studies

If LBDs are associated with loading, then changes in loading should change injury and absenteeism rates. But surprisingly, the literature does not provide clear guidance for reaching this objective; it needs interpretation. The traditional ergonomic approach is to establish a criterion that, if applied in the job design, would lead to a reduction in incidence rate. In a review in 1996, Frank and colleagues suggested that modified work with lower demands can

reduce the number of injuries, but that other changes such as organizational parameters made it difficult to conclude that physical demand changes accounted for any reported differences. This is because very few studies have simply altered job demands. Furthermore, Winkel and Westgaard (1996) noted that the implementation of ergonomics can lead to what they called an ergonomic pitfall. That is, the new ergonomic consciousness causes many workers to report concerns that they did not previously realize were a result of their job situations. Thus, many newly implemented ergonomic intervention programs have resulted in a temporary rise in musculoskeletal disorders. This effect appears to have confounded any study of ergonomic efficacy of insufficient duration. The number of good investigations documenting the effects of job change, or the implementation of ergonomic principles, is also low because such research is very time-consuming and difficult to perform on-site.

Nonetheless, several nice studies qualify by documenting only the effect of ergonomic job redesign. For example, in a report on a series of studies of working women in Norway, Aaras (1994) noted a collectively documented reduction in sick leave due to musculoskeletal troubles as a result of job redesign (specifically, from 5.3 to 3.1%) and a reduction in employee turnover from 30.1 to 7.6%.

Rehab and Prevention Studies

Loisel and colleagues (1997) conducted a very interesting and important study in which they used a randomized control trial design with four groups. One group of people with back injuries received clinical interventions consisting of a visit to a back pain specialist, back school, and functional rehabilitation therapy. Another group received an occupational intervention consisting of a visit to an occupational physician and then an ergonomist to arrive at ergonomic solutions. A third group, the full intervention group, received both of these approaches, and the fourth group received usual care. The group receiving full intervention returned to regular work 2.41 times faster than the usual-care group, although the specific effect of occupational intervention (ergonomics) accounted for the largest proportion of this result with a rate ratio of return to regular work of 1.91.

Those paying for injury (government agencies, compensation boards, and

the insurance industry) could reasonably argue from this evidence that, to reduce costs, care for those with injured backs should be removed from medical hands (once the medics have ruled out red flag conditions such as tumors) and given to ergonomists! Tongue-in-cheek as this statement may be, its point is worthy of deep consideration. Hopefully, the last section of this book will provide clues for more efficacious medical interventions. Krause, Dasinger, and Neuhauser (1998) reviewed the literature on the role of modified work in the return to work and rated as high quality the Loisel study, as well as the study of Baldwin, Johnson, and Butler (1996). This latter review concluded that modified work (involving the modification of musculoskeletal loading) is effective in facilitating the return to work of disabled workers.

In addition to workplace design modifications, workplace interventions may benefit from addressing the personal movement strategies of each worker. In a fascinating study, Snook and colleagues (1998) demonstrated that of 85 patients randomly assigned to a group that controlled the amount of early-morning lumbar flexion, the experimental group had a significant reduction in pain intensity compared to a control group. When the control group received the experimental treatment, they responded with similar reductions. This is yet another example of how personal spine motion patterns and loading posture can influence whether the person will become injured. The next section blends notions of job design with personal movement strategies to reduce loading and the risk of LBD.

Studies on the Connection Between Fitness and Injury Disability

It is fruitful to discuss briefly the role of fitness in the link between injury and disability. Although several studies have shown links between various fitness factors and the incidence of LBD (e.g., Suni et al., 1998, who showed that higher $\dot{V}O_2$ max scores were linked to LBDs), these cross-sectional studies cannot infer causation.

Probably the most widely cited longitudinal study was that of Cady and colleagues (1979), who assessed the fitness of Los Angeles firefighters and noted that those who were rated fitter had fewer subsequent back injuries. However, what is not widely quoted by those citing this study is that when

the fitter firefighters did become injured, the injury was more severe. Perhaps those who were fitter were willing to experience higher physical loads. Several have suggested that a psychological profile is associated with being fit (e.g., Farmer et al., 1988; Hughes, 1984; Ross and Hayes, 1988; Young, 1979) and that the unfit may complain more about more minor aches. Along those lines, some athletes have demonstrated the ability to compete despite injury. Burnett and colleagues (1996) reported cricket bowlers with pars fractures who were still able to compete. Is this due to their supreme fitness and ability to achieve spine stability, or to their mental toughness? Perhaps it is both. The issue remains unresolved.

It is also interesting to note that personal fitness factors appear to play some role in first-time occurrence. Biering-Sorensen (1984) tested 449 men and 479 women for a variety of physical characteristics and showed that those with larger amounts of spine mobility and lower extensor muscle endurance (independent factors) had an increased occurrence of subsequent first-time back troubles. Luoto and colleagues (1995) reached similar conclusions. It would appear that muscle endurance, and not anthropometric variables, is protective.

Our most recent work with firefighters and elite task force police officers adds further insight into the role of fitness (Frost et al., 2011, 2012, 2014, 2015, in press; McGill et al., 2013a, 2013b). From this collective work, we observed that fitter people have more back pain. There are several possible explanations. In athletic groups, we have observed that the fitter ones get more play time and thus more exposure (McGill, Anderson, and Horne, 2012). It is quite possible that the fitter firefighters and police officers take on the challenging physical tasks in a crew, whereas the unfit beg off.

We also strongly suspect that many in these occupational groups have become enamored with the types of warrior training approaches characterized by excessive repetitions of substantial loads until exhaustion in the weight room. Another consideration is that in becoming fitter, their work capacity is expanded. Adding capacity without movement competency appears to lead to pain. We trained two groups of firefighters: one with warrior training approaches of adding more load without concern over movement quality, and another group with coaching to ensure movement competency. The coached group maintained movement quality in occupational exertions requiring pushing, pulling, lifting, chopping, and carrying. The warrior trained group did not. Thus, getting fitter, and by default adding more capacity, needs

movement coaching to reduce injury rates (Frost et al., 2011, 2012, in press).

Beyond Ergonomics: Is It Time to Modify the Worker?

Ergonomics is often described as fitting the task to the person. Many efficacy studies now show that ergonomic job change is effective to varying degrees, but not as effective as might be thought. For example, a study of mechanized lifting aids in a patient care facility showed that the ergonomic aids reduced physical demands and decreased staff fatigue, yet injury rates remained unchanged (Yassi et al., 2001). Although training the worker has been shown to be necessary with virtually any physical ergonomic intervention (e.g., van der Molen et al., 2005), the way the worker moves also appears to be highly important. Reducing injury with ergonomic approaches appears to have some efficacy, but any further advances will probably require changing the person to fit the task. Gagnon (2003) observed expert lifters and concluded that their lifting strategies and personal body movements led to their health success. Our work for the past several years has converged on the notion that the way the exertion is performed determines the risk. Our quantification of martial arts tasks, weightlifting, strongman competitions, and so on, proves that technique determines the risk in very challenging physical tasks. The next section of this chapter incorporates these notions within the context of reducing the risk of back injury.

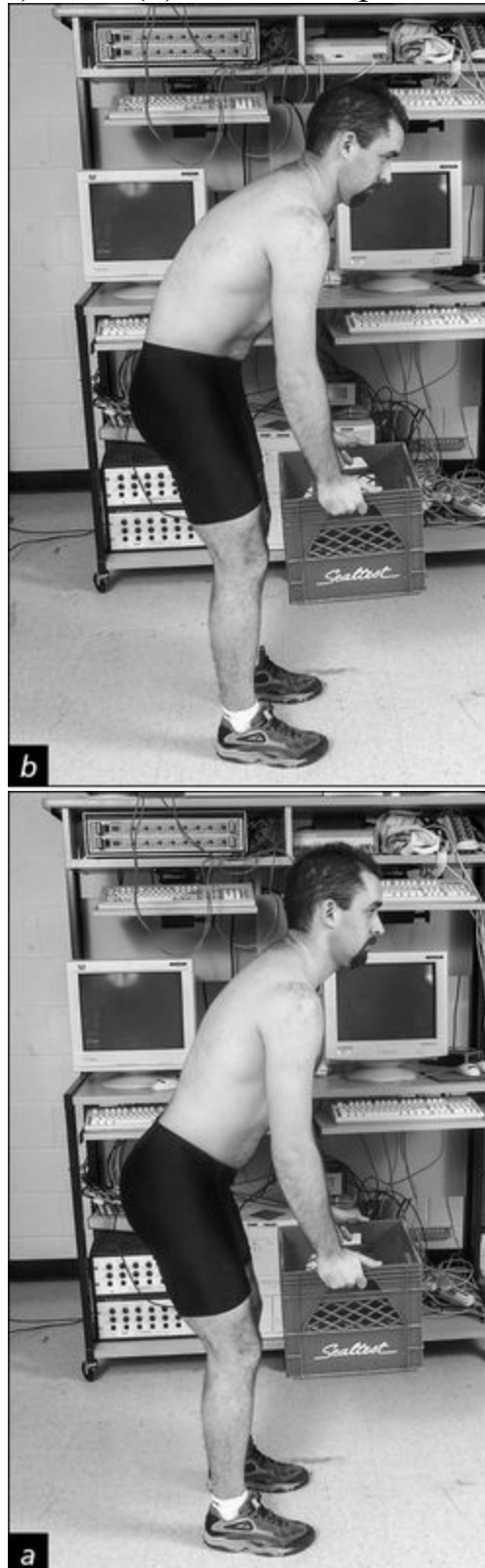
LBD Prevention for Workers

This section addresses a list of issues that are scientifically justifiable to reduce the risk of occupational LBDs.

Should Workers Avoid End Range of Spine Motion During Exertion?

Generally, the answer to whether end range of motion should be avoided is yes—for several reasons. First of all, the strategy works (Ikeda and McGill, 2012; Suni et al., 2013). Generally, if the spine is under load, it is best not to move it—keep it stiff. This principle is not contestable. The second best choice, if the spine is bent during a lift, is to hold it in an isometrically fixed posture with the motion focused about the hip joints. The worst choice is to move the spine over and over, because the combination of load with motion will slowly and cumulatively delaminate the disc collagen. Furthermore, maintaining a more neutrally lordotic spine maximizes shear support, ensures a high tolerance of the joint to withstand compressive forces, eliminates the risk of ligamentous damage because the ligaments remain unstrained, and qualitatively emulates the spine postures that Olympic lifters adopt to avoid injury (see [figure 7.1](#), a and b, for an illustration of a flexed and a neutral spine). Unfortunately, this issue has become confused with issues such as whether it is better to stoop or squat. Another source of confusion has evolved from the common recommendation to perform a pelvic tilt when lifting; the scientific base for this clinically popular notion is nonexistent. Performing a pelvic tilt increases tissue stresses and the risk of injury!

Figure 7.1 (a) Flexing the torso involves either hip flexion or spine flexion, or (as in this case) both. (b) A neutral spine with hip flexion.



Photos from Stuart McGill

What spine and hip posture best minimizes the risk of injury? We do know that very few lifting tasks in industry can be accomplished by bending the knees and not the back. Furthermore, most workers rarely adhere to this technique when repetitive lifts are required—a fact that is quite probably due to the increased physiological cost of squatting compared with stooping (Garg and Herrin, 1979). However, a case can be made for preserving neutral lumbar spine curvature during lifting (specifically, avoiding the end range of spine motion about any of the three axes). This is a different concept from trunk angle, because the posture of the lumbar spine can be maintained independent of thigh and trunk angles.

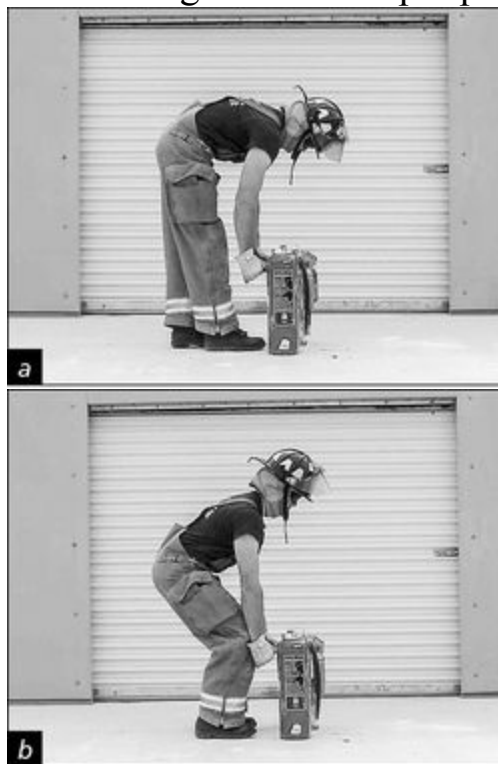
The literature is confused between trunk angle or inclination and the amount of flexion in the lumbar spine. Bending over is accomplished by either hip flexion or spine flexion, or both. It is the issue of specific lumbar spine flexion that is of importance here. Normal lordosis can be considered to be the curvature of the lumbar spine associated with the upright standing posture. (To be precise, the lumbar spine is slightly extended from elastic equilibrium during standing; see Scannell and McGill, 2003.) In [figure 7.2](#), a warehouse worker is sparing his spine by avoiding end range of spine motion even though he is not bending the knees; he has accomplished torso flexion by rotating about the hips. In [figure 7.3](#), a firefighter is demonstrating a spine-damaging technique ([figure 7.3a](#)) versus a spine-sparing technique ([figure 7.3b](#)) for lifting extinguishers. Chapter 8 offers further occupation-specific examples (e.g., the construction workers demonstrating spine-sparing postures in [figure 8.3](#)).

Figure 7.2 This warehouse worker is not bending the knees, yet he is sparing the spine by electing to bend and rotate about the hips; the lumbar spine is not flexed.



Photos from Stuart McGill

Figure 7.3 (a) This firefighter, who is flexing his lumbar region, is loading the passive tissues and increasing his risk of back troubles. (b) Avoiding full lumbar flexion and rotating about the hips spare the spine.



Photos from Stuart McGill

Chapter 4 addresses the load distribution among the tissues of the low back. Of interest here are the dramatic effects on shear loading as a function of spine curvature. Recall the following facts:

- A spine that is not fully flexed ensures that the pars lumborum fibers of the longissimus thoracis and iliocostalis lumborum can provide a supporting posterior shear force on the superior vertebra, whereas full flexion causes the interspinous ligament complex to strain, imposing an anterior shear force on the superior vertebra. For these reasons, avoiding full flexion not only ensures a lower shear load but also eliminates ligament damage.
- A fully flexed spine is significantly compromised in its ability to withstand compressive load.
- Because herniation of the nucleus through the annulus is caused by repeated or prolonged full flexion, avoidance of the posture minimizes the risk of herniation (a cumulative trauma problem) and minimizes the

stresses on any developing annular bulges.

Olympic lifters provide a convincing example of the efficacy of avoiding lumbar flexion in lifting. They lock their spines in a neutral posture and emphasize rotation about the hips (see [figure 7.4](#)), lifting enormous weights without injury. The ambulance attendants in [figure 7.5](#) are also sparing their backs by locking the lumbar regions to avoid flexion.

Figure 7.4 World-class performances in Olympic weightlifting are characterized by a lumbar spine locked in neutral and the motion taking place about the hips and knees. Spine motion would lead to either injury or an uncompetitive lift.



Figure 7.5 These ambulance attendants are attempting to spare their backs by avoiding lumbar flexion and lifting together to share the load. They have also been taught to lightly contract the stabilizing abdominal musculature.

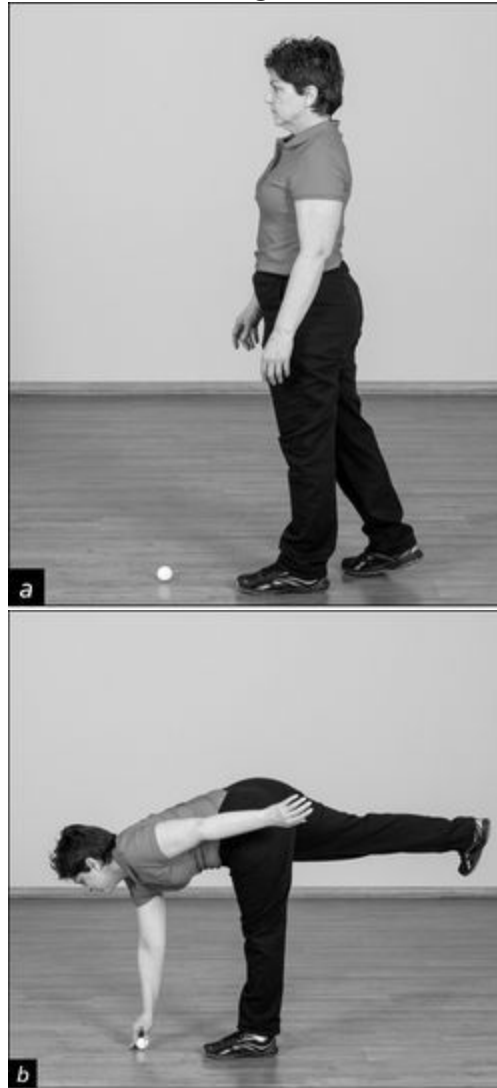


Photos from Stuart McGill

Thus, the important issue is not whether it is better to stoop lift or to squat lift; rather, the emphasis should be to place the load close to the body to reduce the reaction moment (and the subsequent extensor forces and resultant compressive joint loading) and to avoid a fully flexed spine. Sometimes it may be better to squat to achieve this; if the object is too large to fit between the knees, however, it may be better to stoop, flexing at the hips but always avoiding full lumbar flexion to minimize posterior ligamentous involvement. (For a more comprehensive discussion, see McGill and Kippers, 1994; McGill and Norman, 1987; Potvin, Norman, and McGill, 1991.)

Yet another spine-sparing lifting posture is the golfer's lift, which reduces spine loads for repeated lifting of light objects (see [figure 7.6](#), a and b) (Kinney, Callaghan, and McGill, 1996). The hips act as a fulcrum in which one leg is cantilevered behind with isometric muscle contraction, forming a counterweight to rotate the upper body back to upright.

Figure 7.6 (a, b) The golfer's lift has been documented to minimize low back motion and reduce the loads on the lumbar tissues by using the leg, which is cantilevered behind, as a counterweight; (c) the hips act as a fulcrum to raise the torso to upright. This is an effective technique for repeated lifting of light objects from floor level. Most still adhere to the general instruction to bend the knees and keep the back straight, not considering the spine-conserving benefits of the golfer's lift.





What Are the Ways to Reduce the Reaction Moment?

A popular instruction is to hold the load close to the torso when handling material. This is biomechanically wise; a reduced lever arm of the load requires lower internal tissue loads to support the reaction moment. But phrasing the principle in terms of holding the load close restricts the notion to lifting tasks. The real biomechanical principle is to reduce the reaction moment. When phrased this way, the principle is now applicable to any task involving the exertion of external force.

One technique is to expand the concept used by archers—termed the archer's bow by some of my Australian colleagues ([figure 7.7](#)). Here, one arm pushes to support the reaction moment needed to support the upper body while the other arm lifts the load.

Figure 7.7 (a) The archer's bow (b) is translated to a lifting situation to minimize back loading. Opposing hand forces are shown with arrows.



Photos from Stuart McGill

Directing the Pushing Force Vector Through the Spine

When pushing a cart handle, directing the pushing force vector through the lumbar spine reduces the reaction moment and therefore the tissue loads and spine load (see [figure 7.8](#)). In contrast, a pushing force directed through

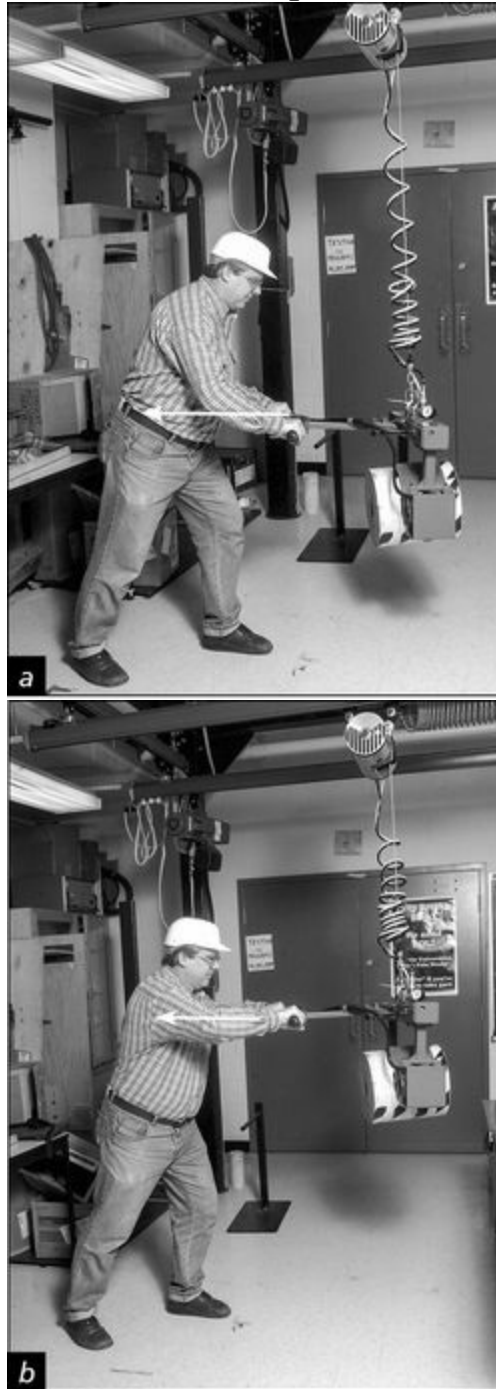
the shoulder would not be directed through the low back (see [figure 7.9](#), a and b); this forms a transmissible vector. The large perpendicular distance from this force to the spine causes a high reaction torque that is balanced by muscular force; this imposes a corresponding compressive penalty on the spine. Redirecting the transmissible vector through the low back reduces this moment arm (and the moment), the muscle forces, and the spine compressive load. Another example from everyday activities is the technique used to open a door; directing the pulling force through the low back is sparing (see [figure 7.10](#), a-c). Another example is vacuuming, which is often reported by our patients to exacerbate their symptoms. Holding the handle to the side creates a large moment arm for the pushing and pulling forces; this heavily loads the back (see [figure 7.11a](#)). Directing the pushing and pulling forces through the low back minimizes the moment arm (see [figure 7.11b](#)) and removes the loads that ensure that patients remain patients! These examples demonstrate how this powerful but rarely practiced technique of skilled control of the transmissible vector spares the back.

Figure 7.8 Pushing through the hands but directing the force vector to pass through the low back minimizes the low back moment (and minimizes the muscular loads).



Photos from Stuart McGill

Figure 7.9 (a) Pushing or pulling forces that pass by the spine with a large moment arm ensure a high moment and corresponding high muscle forces and spine loads. (b) Pushing forces directed through the low back minimize the moment and spine load.



Photos from Stuart McGill

Figure 7.10 (a) Opening a door by directing hand forces through the low back spares the spine. Many people are not taught how to optimize this principle during the performance of daily tasks. (b) Instead, they sacrifice the spine by opening the door so that the force is lateral to the lumbar spine creating a twisting torque or (c) produce a pulling force that passes over the low back.





Photos from Stuart McGill

Figure 7.11 (a) Vacuuming is often reported as problematic by people with bad backs. The transmissible vector in this example has a large moment arm with respect to the back and requires large twisting torque forces (causing pain). (b) The transmissible vector is directed through the low back, minimizing the load and the pain and enabling vacuuming.





Photos from Stuart McGill

Diverting the Force Around the Lumbar Spine

Still other very effective ways exist to reduce the reaction moment. The next examples demonstrate how workers skillfully spare the low back by diverting force around the lumbar region when lifting. The workers lifting the shaft in a paper factory in [figure 7.12](#) are minimizing the reaction moment by lifting the shaft with the thigh (by plantar flexing the foot). In this way, the weight of the shaft is directed down the thigh to the floor, bypassing the upper-body linkage and spine. Prior to redesigning the job, the workers had to lift the shaft with arm forces, causing low back problems and motivating our involvement. [Figure 7.13](#), a through c, illustrates how a worker can perform the same maneuver to lift a patient. Observe that the lifter is pulling the patient's pelvis onto his thigh and standing along with the patient. (Note that patient lifts cannot be performed with universal technique because patients differ in many ways, including their ability to offer assistance.) Minimal forces are transmitted down the spine. Shoveling snow by resting the forearm on the thigh involves the same spine-sparing principle (see [figure](#)

[7.14](#), a and b).

Figure 7.12 This task requires a worker to lift a heavy shaft while another worker slides on a core from the end. A high incidence of back troubles motivated our consulting recommendation to install a bar for the foot, which is plantar flexed to raise the shaft now supported on the thigh. The forces now transmit directly down the leg, completely bypassing the arm and spine linkage.



Photos from Stuart McGill

Figure 7.13 (a) Lifting the patient can be performed with minimal loading down the spine provided the height of the seat is close to the height of the lifter's thigh when in a squat position. (b) The patient's pelvis is pulled and slid onto the lifter's thigh, and (c) then the lifter stands up, hugging the patient's pelvis and minimizing the forces up the arms and down the spine. This spine-sparing technique is reserved for patients who are able to stand.





Figure 7.14 (a) Shoveling snow with a large moment arm for the load on the shovel loads the back. (b) Resting the arm on the thigh directs the forces to the ground, bypassing the arm and spine linkage.



Photos from Stuart McGill

A Note on Pushing and Pulling

Our studies comparing the expert techniques of firefighters to the techniques of graduate students were noted in chapter 4 (Lett and McGill, 2006). Specifically, for tasks such as hose pulling, the firefighters braced the body and pulled the hose with the force vector traveling through the low back

to minimize lumbar loading (see [figure 7.15](#)). Some of our recent work documenting the techniques employed by hospital workers when they transfer patients highlighted once again this important technique (McGill and Kavcic, 2005). On occasion, friction-reducing devices are placed underneath patients that need to be transferred. Providers can then apply pushing and pulling forces to the patient. Even with the friction-reducing devices, the principle of directing the hand forces through the lumbar spine is a very important component in minimizing spine forces.

Figure 7.15 Directing pulling forces through the low back helps to minimize spine forces. Spine posture (neutral) and whole-body posture (to reduce the reaction moment) are also important variables for injury avoidance.



Photos from Stuart McGill

Reducing the Load

Finally, workers can use skill to reduce the actual load and hand forces when lifting large objects. Some tasks can be performed by lifting only half of the object at a time. For example, when loading long logs onto the back of a truck, a worker could lift just one end (effectively handling only half of the full weight of the log) and place it on the bed. The worker could then walk around to the other end of the log and lift while sliding the log onto the bed (see [figure 7.16](#)). Each lift is half of the total load. A sequence of unloading a refrigerator demonstrates the same technique; the full weight of the object is never lifted (see [figure 7.17](#), a-f)! The worker lifting the mini-refrigerator (shown in [figure 7.18](#), a and b) tilts it up onto an edge, raising its center of gravity together with the initial lifting height. Lifting from this higher starting position reduces the necessary moment. The concept of minimizing the reaction moment is much more robust than simply telling people to hold the load close to the body.

Figure 7.16 This girl loads the log into the truck by lifting only half its weight at a time. First she lifts one end onto the truck. Then she lifts the other end and slides the log onto the bed.



Photos from Stuart McGill

Figure 7.17 This worker unloads the refrigerator from the trailer without having to lift its full weight. (a) He “walks” the refrigerator on its corners to the edge of the trailer, where he balances it over the lip of the bed. (b) Next, he slides the refrigerator down the trailer’s tailgate. (c) He walks the refrigerator clear of the trailer and leaves it standing upright while he retrieves the dolly. (d, e) He pushes one edge of the refrigerator up just enough to slide the dolly under it. (f) Finally, he and a coworker wheel the appliance away.





Photos from Stuart McGill

Figure 7.18 (a) This man lifts the mini-refrigerator by tilting it up onto an edge, raising its center of gravity together with the initial lifting height. (b) Lifting from this higher starting position reduces the necessary moment so the man can reach the standing position without damaging his back.



Should One Avoid Exertion Immediately After Prolonged Flexion?

Recall that prolonged flexion causes both ligamentous creep and a redistribution of the nucleus within the annulus (see chapter 4). In this way the spine tissues have a loading memory. Further, evidence from Jackson and colleagues (2001) and Le and colleagues (2009) suggests that prolonged flexion modifies the extensor neurological response and causes muscle spasming, at least until the ligamentous creep has been restored. In this case, it was 7 hours! Spine stability would be compromised during this period, and the risk of annulus damage remains temporarily high. If possible, after prolonged stooping or sitting activities, people should spend time standing upright before attempting more strenuous exertions. For example, it would be very unwise for the gardener in [figure 7.19](#) to stand and lift a bag of peat moss or for the shipper/receiver in [figure 7.20](#) to rise and immediately begin loading pallets. How long should one wait before performing an exertion? Data from Twomey and Taylor (1982) derived from cadaveric spines suggest that age delays the recovery of the spinal tissues. McGill and Brown (1992) noted that residual laxity remained in the passive tissues even after a half hour of standing following prolonged flexion—although the flexion was extreme in an occupational context. At least 50% of the joint stiffness returned after 2 minutes of standing following the session of prolonged flexion.

Figure 7.19 This gardener would be unwise to stand and immediately lift a heavy object after spending a long period of time in a stooped posture. Rather, standing for a brief period and even extending will prepare the disc and posterior passive tissues and reduce the risk of injury.



Photos from Stuart McGill

Figure 7.20 This shipper/receiver's work is characterized by periods of sitting followed immediately by lifting when a truck pulls up to the loading dock. Again, standing and waiting for a few minutes prior to the exertions will reduce the risk of injury.



Photos from Stuart McGill

As noted in chapter 4, because the mechanics of the joints are modulated by previous loading history, one should never move immediately to a lifting task from a stooped posture or after prolonged sitting. Rather, one should first stand, or even consciously extend the spine, for a short period. Obviously, people in certain occupations such as emergency ambulance personnel cannot follow this guideline. They arrive at an accident scene without the luxury of time to warm up their backs and may have to perform nasty lifts—such as a 150 kg (331 lb) heart attack victim out of a bathtub! For these people, the only strategy is to avoid a fully flexed spine posture while driving so that the spine is best prepared to withstand the imposed loads. This may be done with accentuated lumbar pads in ambulance seats and with worker training.

Should Intra-Abdominal Pressure (IAP) Be Increased During Lifting?

Generally, for occupational applications, the answer is no: at least IAP should not be increased consciously. Recall the discussion in chapter 5

concluding that IAP does not reduce spine loading but does act to stiffen it against buckling. By completing the rehabilitation training advocated in the final section of this book, people can train their breathing and IAP to be independent of exertion. In this way any specific instructions regarding breathing and exertion become moot points. In most cases IAP will rise naturally, and no further conscious effort is required.

A final caveat is required here. Very strenuous lifts, if they must be performed, require the buildup of IAP to increase torso stiffness and ensure stability (Cholewicki, Juluru, and McGill, 1999). On the other hand, we know that a substantially increased intrathoracic pressure (as occurs with lifting) compromises venous return (Mantysaari, Antila, and Peltonen, 1984). Further, breath holding during exertion raises both systolic and diastolic blood pressure (Haslam et al., 1988), which can be a concern for some. Blackout is not uncommon in strenuous lifting even though it is not clear which mechanism is responsible (MacDougall et al., 1985). Reitan (1941) proposed that blackout may be due to elevated central nervous system fluid pressure (IAP also raises central nervous system fluid pressure in the spine and up to the head), whereas Hamilton, Woodbury, and Harper (1944) proposed that an increase in cerebrospinal fluid pressure might actually serve a protective function (i.e., the consequent decrease in transmural pressure across the cerebral vessels could actually decrease the risk of vascular damage). At this point, given these issues and a lack of full understanding, moderate IAP may be warranted with the understanding of the negative side effects. Extreme lifting efforts involving conscious increases in IAP should not be performed at work. These situations are discussed later.

Are Twisting and Twisting Lifts Particularly Dangerous?

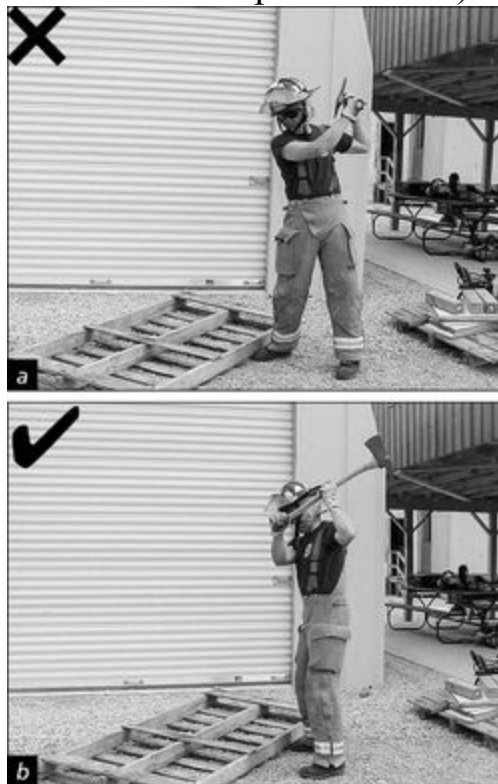
Although several researchers have identified twisting of the trunk as a factor in the incidence of occupational low back pain (Frymoyer et al., 1983; Troup, Martin, and Lloyd, 1981), the mechanisms of risk require some explanation. The kinematic act of twisting has been confused in the literature with the kinetic variable of generating twisting torque. Twisting torque in the torso can be accomplished whether the spine is twisted or not.

Generally, twisting to moderate degrees without high twisting torque is

not dangerous. Some have hypothesized based on an inertia argument that twisting quickly imposed dangerous axial torques upon braking the axial rotation of the trunk at the end range of motion. Farfan and colleagues (1970) proposed that twisting of the disc is the only way to damage the collagenous fibers in the annulus leading to failure. They reported that distortions of the neural arch permitted such injurious rotations. Shirazi-Adl, Ahmed, and Shrivastava (1986) conducted more detailed analyses of the annulus under twist. They supported Farfan's contention that twisting indeed can damage the annulus at end range, but also noted that twisting is not the sole mechanism of annulus failure. In contrast, some research has suggested that twisting in vivo is not dangerous to the disc because the facet in compression forms a mechanical stop to rotation well before the elastic limit of the disc is reached; thus, the facet is the first structure to sustain torsional failure (Adams and Hutton, 1981). In a study of ligament involvement during twisting, Ueno and Liu (1987) concluded that the ligaments were under only negligible strain during a full physiological twist. However, an analysis of the L4-L5 joint by McGill and Hoodless (1990) suggested that posterior ligaments may become involved if the joint is fully flexed prior to twisting.

Generating twisting torque is a different issue (see [figure 7.21](#), a and b). Because no muscle has a primary vector direction designed to create twisting torque, all muscles are activated in a state of great cocontraction. This results in a dramatic increase in compressive load on the spine when compared to an equivalent torque about another axis. For example, data from a combination of our previous studies indicate that supporting 50 Nm in the extension axis imposes about 800 N of spinal compression. The same 50 Nm in the lateral bend axis results in about 1,400 N of lumbar compression, but 50 Nm in the axial twist axis would impose over 3,000 N (McGill, 1997). It appears that the joint pays dearly to support even small axial torques when extending during the lifting of a load.

Figure 7.21 (a) Twisting the torso is occurring at the same time that twisting torque is required, which is a dangerous combination. (b) Generating the twisting torque but restricting the torso twist is a spine-sparing strategy (and can also enhance performance).



Photos from Stuart McGill

Although creating twisting torque increases the load on the spine, being twisted reduces the ability of the spine to bear load (Aultman et al., 2004, and Drake et al., 2005, observed the loss in load tolerance). Marshall and McGill (2010) documented how cyclic twisting causes circumferential delamination of the collagen, or what are known radiologically as disc tears. Thus twisting, when combined with twisting torque, becomes problematic.

To conclude, the generation of axial twisting torque when the spine is untwisted does not appear to be of particular concern. Nor is the act of twisting over a moderate range without accompanying twisting torque. But generating high torque while the spine is twisted appears to create a problematic combination and a high risk. This is of particular concern in several sports and is addressed in that context in chapter 11.

Is Lifting Smoothly and Not Jerking the Load Always

Best?

The answer to this question is no. We have all heard that a load should be lifted smoothly and not jerked. This recommendation was most likely made based on the belief that accelerating a load upward increases its effective mass by virtue of an additional inertial force acting downward together with the gravitational vector. However, this may not always be the case. It is possible to lift a load by transferring momentum from an already moving segment. Autier, Lortie, and Gagnon (1996) showed that, compared to novice lifters, expert materials handlers sometimes choose techniques that make more efficient use of momentum transfer. They do not always lift slowly and smoothly.

Troup and Chapman (1969) referred to the concept of momentum transfer during lifting, as did Grieve (1975), who coined the term kinetic lift. Later, McGill and Norman (1985) documented that smaller low back moments were possible in certain cases using a skillful transfer of momentum. For example, if a load is awkwardly placed, perhaps on a work table 75 cm (30 in.) from the worker, a slow, smooth lift would necessitate the generation of a large lumbar extensor torque for a lengthy duration of time—a situation that is most strenuous on the back. However, the worker could lift this load with a very low lumbar extensor moment or quite possibly no moment at all. If the worker leaned forward and placed his hands on the load, with bent elbows, the elbow extensors and shoulder musculature could thrust upward, initiating upward motion of the trunk to create both linear and angular momentum in the upper body (note that the load has not yet moved). As the worker straightens his arms, coupling takes place between the load and the large trunk mass (as the hands then start to apply upward force on the load), transferring some, or all, of the body momentum to the load and causing it to be lifted with a jerk.

This mechanical solution was proven to be effective in a very early experiment in my research career. The person shown in [figure 7.22](#) is demonstrating a task in which the load can be lifted slowly, which would load the low back unduly, or with the kinetic lift technique, which, if correctly performed, will spare the back. (Obviously the markers on the model's body were to assist the measurement of body segment movement.) This highly skilled inertial technique is observed quite frequently throughout industry and in some athletic events such as competitive weightlifting, but it